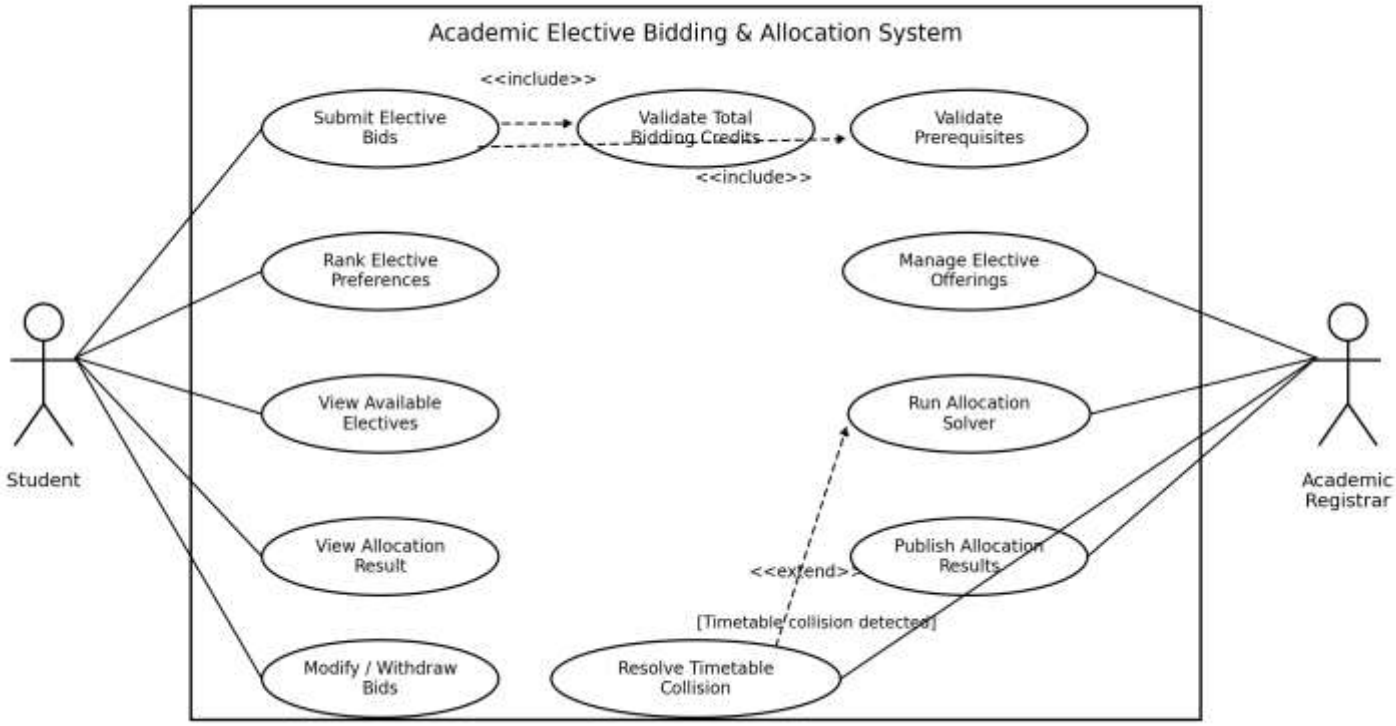
Academic Elective Bidding & Allocation System

1. Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow students to distribute a maximum of 100 bidding credits across ranked elective preferences and validate prerequisite course completion.	High	A student can distribute credits across eligible electives, and the system rejects any bid where the total exceeds 100 or a prerequisite is not satisfied.	Ensures fair bidding and prevents ineligible elective selections.
FR-002	Functional	The system shall allow students to view available electives along with prerequisite, seat-capacity, and timetable information before submitting bids.	High	The student can view the current elective list and relevant course details before placing bids.	Helps students make informed elective choices.
FR-003	Functional	The system shall allow students to rank elective preferences and modify or withdraw their bids before the bidding deadline.	Medium	Changes made before the deadline are saved and reflected in the final bid submission; changes after the deadline are rejected.	Allows students to revise preferences while preserving a clear submission deadline.
FR-004	Functional	The system shall allow the Academic Registrar to configure elective offerings, seat capacities, prerequisites, and bidding deadlines.	High	The registrar can create or update elective settings, and the latest valid settings are used by the bidding and allocation processes.	Provides administrative control over each elective allocation cycle.
FR-005	Functional	The system shall allocate electives using submitted bids while respecting seat limits, prerequisite eligibility, ranked preferences, and timetable collision constraints.	High	The solver produces an allocation in which no student is assigned to an ineligible course, no course exceeds capacity, and timetable collisions are prevented.	Ensures consistent and constraint-aware elective allocation.
NFR-001	Non-Functional - Performance	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	During performance testing with 5,000 student bids, the allocation and conflict-resolution process completes in less than 30 seconds.	Ensures results can be generated quickly even at university scale.
NFR-002	Non-Functional - Security	Only authenticated students and authorized Academic Registrar users shall access bidding and allocation functions according to their assigned roles.	High	Students cannot access registrar-only controls, unauthorized users cannot submit or modify bids, and restricted requests are rejected.	Protects bidding data and prevents unauthorized allocation changes.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: Submit Elective Bids

Use Case ID	UC-01
Use Case Name	Submit Elective Bids
Primary Actor	Student
Supporting Actors	Academic Registrar

Preconditions

1. The student has a valid account and is logged into the system.
2. The elective bidding window is open.
3. The Academic Registrar has configured the available electives, seat capacities, prerequisites, and timetable details.
4. The student has access to the academic records required for prerequisite validation.

Postconditions

On Success:

- The student's ranked elective preferences and bidding-credit distribution are stored.
- The total allocated bidding credits do not exceed 100.
- All submitted elective preferences satisfy the required prerequisites.
- The valid bid is available to the final allocation solver.

On Failure:

- The invalid bid is not accepted.

- The student is shown the reason for rejection, such as exceeding 100 credits or failing a prerequisite.
- The student may correct the bid and resubmit while the bidding window remains open.

Main Success Scenario

1. The Student logs into the Academic Elective Bidding & Allocation System.
2. The Student selects the elective bidding option.
3. The system displays the available electives with prerequisite and timetable information.
4. The Student reviews the available electives.
5. The Student ranks the preferred electives.
6. The Student distributes bidding credits across the ranked preferences.
7. The Student submits the elective bid.
8. The system checks that the total allocated bidding credits do not exceed 100.
9. The system validates prerequisite course completion for each selected elective.
10. The system validates that the bid was submitted within the bidding window.
11. The system accepts the valid bid.
12. The system stores the student's ranked preferences and bidding-credit distribution.
13. The system confirms successful submission to the Student.
14. After the bidding period ends, the valid bid becomes available to the final allocation solver.

Alternate Flow

A1 - Invalid Bid Due to Credit Limit or Prerequisite Failure

1. The Student submits the elective bid.
2. The system detects that the total bidding credits exceed 100 or that a selected elective has an unmet prerequisite.
3. The system rejects the bid.
4. The system displays the specific validation error to the Student.
5. The Student adjusts the credit distribution or removes the ineligible elective.
6. The Student resubmits the corrected bid before the bidding deadline.
7. The system validates the corrected bid again.
8. If all validations pass, the flow continues from Step 11 of the Main Success Scenario.